



Brachial-ankle pulse wave velocity in healthy Japanese adolescents: reference values for the assessment of arterial stiffness and cardiovascular risk profiles

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Abstract

Brachial-ankle pulse wave velocity (baPWV) has been used as a simple and convenient method of evaluating arterial stiffness and is considered useful for screening subclinical vascular damage in primary care settings and in large populations. The aim of this study was to determine the reference values of baPWV in adolescents based on age and sex to evaluate and classify vascular abnormalities in this age population. Noninvasive baPWV measurements using an oscillometric device were carried out in 5936 participants aged 12–18 years who had no history of syndromic obesity, hypertension, dyslipidemia, diabetes mellitus, or renal disease. Of these, we extracted data from 4524 healthy adolescents free from atherosclerotic risk factors and constructed sex-specific reference percentiles normalized to age using the LMS method. The baPWV increased with age and was significantly higher in males than in females. Multivariate regression analysis demonstrated that age, body mass index, and blood pressure were the major determinants of baPWV for both males and females. Among a sample of 3825 adolescents, including individuals with risk factors, the prevalence of central obesity, raised blood pressure, raised triglycerides, reduced high-density lipoprotein cholesterol, and impaired fasting glycemia significantly increased along with the standardized baPWV *z*-score level. In addition, there was a graded nature regarding the association between the baPWV *z*-score and the clustering number of the metabolic syndrome components. These results suggest that the proposed reference values of baPWV could help classify vascular abnormalities of adolescents caused by the presence of risk factors and thereby contribute to determining individuals with cardiovascular risk in this age population.

Keywords Adolescents · Age · Atherosclerotic risk factors · Brachial-ankle pulse wave velocity · Sex

Introduction

It has been reported that, as early as the second decade of life, the arterial wall structure and function can be influenced by atherosclerotic risk factors [1–4]. Vascular changes may not be confined to individuals with conditions such as severe obesity, hypertension, dyslipidemia, or diabetes mellitus. Furthermore, the presence of subclinical vascular

damage in childhood and adolescence frequently leads to the early onset of cardiovascular disease in adulthood [5, 6]. These facts emphasize the need for the evaluation of arterial wall properties early in life to identify individuals with an increased risk of cardiovascular events within the initial phase of vascular aging.

Arterial stiffness reflects systemic atherosclerosis at various sites in the vascular tree [7], and it is believed to play a role in the development of cardiovascular disease [8, 9]. The measurement of brachial-ankle pulse wave velocity (baPWV) has been used as a simple and convenient method for evaluating arterial stiffness and is considered useful for screening subclinical vascular damage in primary care settings and in large populations [10, 11]. Previous studies in adults revealed that baPWV is closely related to the extent of atherosclerosis and is increased by the presence of various risk factors [11–16]. In addition, baPWV has been reported to be a useful

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surrogate marker for the risk prediction of future onset of cardiovascular disease [17]. The number of studies using baPWV to evaluate arterial stiffness in the pediatric population has rapidly increased. Most studies showed that a higher baPWV was related to obesity, elevated BP, lipid and glucose tolerance abnormalities, and total homocysteine levels in a group of healthy adolescents [2, 18–20], whereas other studies in children and adolescents of a wide age range could not demonstrate an association between baPWV and adiposity parameters [21, 22]. The conflicting results in the previous reports can be attributed partially to whether the effects of growth-related physiological changes in baPWV are adequately considered. As the baPWV of adolescents is greatly influenced by age and sex [2, 18, 20, 22], the impact of atherosclerotic risk factors on baPWV might be masked by the effect of these factors. Therefore, although baPWV measurements may be useful for evaluating arterial wall properties in younger populations, some adjustments related to age and sex are required to estimate the early vascular changes that might be caused by the presence of risk factors. However, for children and adolescents, there are very few reports demonstrating the physiological values of baPWV related to age and sex [20, 23], and this has limited its usage in clinical practice and population studies.

The aim of this study was to investigate baPWV in healthy adolescents aged 12–18 years and to provide reference values based on age and sex that could contribute to the classification of vascular abnormalities and the identification of individuals with cardiovascular risk in this age population.

Methods

Study participants

The study participants consisted of the adolescent general population in a school-based setting. We conducted health-screening examinations between 2002 and 2018 in students in a total of 11 different public schools (1 elementary school, 6 junior high schools, and 4 high schools) located in Osaka, Hyogo, and Wakayama, Japan. Recruitment for participation in the examination together with an explanation of the aims and procedures was performed by school health staff at each school. Of the participants, 5936 adolescents aged between 12 and 18 years were enrolled in this study. Individuals were included if they had no history of treatment for syndromic obesity, hypertension, diabetes mellitus, dyslipidemia, or renal disease and no electrocardiographic evidence of coronary disease or cardiac arrhythmia.

The study protocol was approved by the Ethics Committee of Wakayama Medical University (No. 596). The

participants and their guardians provided informed consent or assent as appropriate after explaining the aims of the study and experimental procedures. The study was performed in accordance with the principles of the Declaration of Helsinki.

Anthropometric and clinical variables

Body height was measured with a portable stadiometer, and weight was measured with a calibrated digital scale. Body mass index (BMI) was calculated as the ratio of weight to height squared (kg/m^2). The BMI *z*-score of each participant was calculated based on the reference values proposed by the World Health Organization (WHO) [24], which provides international BMI cutoff points for overweight and obesity in children and adolescents according to age and sex. Body fat mass was estimated from total body resistance using a bioelectrical impedance analyzer (TBF-310; TANITA Co., Tokyo, Japan). Waist circumference (WC) was measured at the midpoint between the iliac crest and the lower edge of the ribs in the midaxillary line using a standard nonstretchable flexible measuring tape. The anthropometric measurements were collected with each participant wearing light clothes without shoes.

After each participant rested in a temperature-controlled and quiet room for at least 5 min, brachial arterial blood pressure (BP) and heart rate were measured 3 times in the left arm using a digital BP monitor (HEM907; OMRON Co., Tokyo, Japan) with an appropriate cuff size (3 cuff sizes: large, 32–42 cm; medium, 22–32 cm; and small, 17–22 cm). The measurements were performed when the participants were in a comfortable supine position to minimize tension or anxiety as much as possible. An average of the last two readings was used to derive the representative values for systolic BP (SBP), diastolic BP (DBP), and heart rate. Pulse pressure was calculated as the difference between the SBP and DBP. Blood samples were obtained from the antecubital vein using standard techniques for the measurement of serum concentrations of total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), triglycerides, fasting blood glucose (FBG), and hemoglobin A1c (HbA1c).

All the measurements were performed during a regular school day and in the morning after an overnight fast for more than 8 h and abstinence from beverages containing caffeine for at least 3 h. The blood sampling was carried out at the beginning of the measurements, and the participants took light food and water after collecting their blood samples.

Pulse wave velocity

Following the BP measurement, participants maintained their supine position and baPWV was measured using a

volume-plethysmographic apparatus (BP-203RPE II/III form PWV/ABI: Omron Colin Co. Ltd., Tokyo, Japan). Occlusion cuffs connected to both plethysmographic and oscillometric sensors were attached to both the upper arms and ankles of the participants. The cuff was selected according to the size of the arm and ankle circumferences of each individual following the manufacturer's recommendations (3 arm cuff sizes: large, 30–38 cm; medium, 20–32 cm; and small, 16–25 cm; and one ankle cuff size: 16–33 cm). Measurements obtained with this apparatus have been shown to have good reproducibility and to be highly correlated with the aortic PWV measured using a catheter-tipped manometer [10]. Using this apparatus, waveform data were obtained from a volume-plethysmographic sensor placed in occlusion cuffs on both the upper arms and the ankles of the participants, and the time intervals (ΔT_{ba}) between the wave front of the brachial waveform and that of the ankle waveform were calculated. The path lengths from the suprasternal notch to the brachium (L_b) and ankle (L_a) were obtained automatically according to the height of the individual. The baPWV was calculated using the following equation: $baPWV = (L_a - L_b) / \Delta T_{ba}$. The mean of the right and left baPWV was used for analysis in this study because a highly significant positive correlation was found between the left and right baPWV ($r = 0.94$, $P < 0.001$).

Statistical analysis

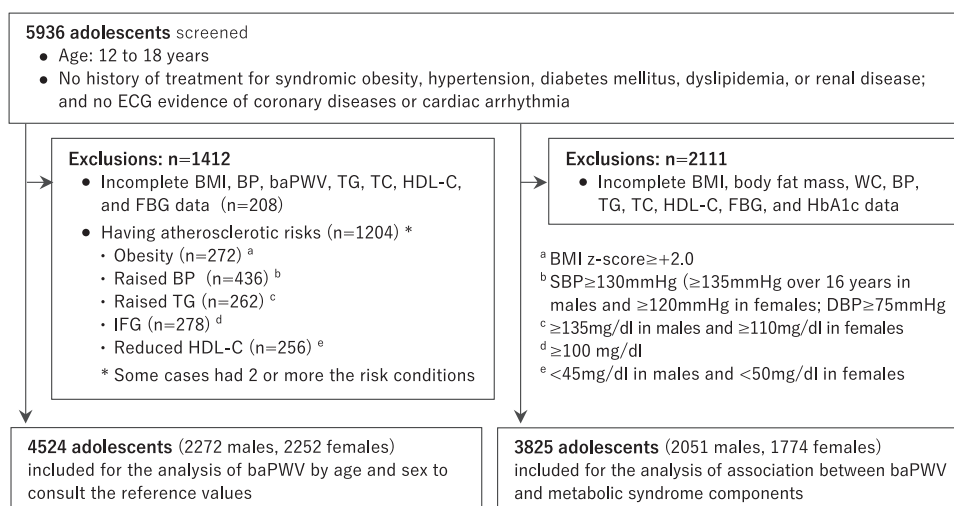
To construct reference values for baPWV, we excluded from the present analysis any adolescent with incomplete baPWV, anthropometric, or clinical measurement data ($n = 208$) or any adolescent with any one of the following atherosclerotic risk factors ($n = 1204$): obesity (BMI z -score $\geq +2.0$); raised BP, raised triglycerides, or impaired fasting glycemia (≥ 95 th percentile predicted for age and sex in study population); or reduced HDL-C (< 5 th percentile)

(Fig. 1). The cutoff values dichotomizing the individuals were as follows: in males, SBP ≥ 130 mmHg (≥ 135 mmHg over 16 years of age), DBP ≥ 75 mmHg, triglycerides ≥ 135 mg/dl, HDL-C < 45 mg/dl, and FBG ≥ 100 mg/dl; in females, SBP ≥ 120 mmHg, DBP ≥ 75 mmHg, triglycerides ≥ 110 mg/dl, HDL-C < 50 mg/dl, and FBG ≥ 100 mg/dl. The BMI z -score was calculated from WHO growth charts [23]. After exclusions, 4524 healthy adolescents (2272 males and 2252 females; mean age, 14.9 ± 1.7 years) were included for the analysis of baPWV by age and sex.

baPWV by age and sex was generated using the LMS method [25]. With this method, the changing distribution based on age is presented as the median (M), the coefficient of variation (S , i.e., the relation between SD and the mean), and a measure for the skewness (L). Using penalized likelihood, the curves for L , M , and S can be fitted as cubic splines by nonlinear regression. The L , M , and S were analyzed for each half-year of age (e.g., age 12.0 includes 12.00–12.49). The points on each percentile curve are calculated by the equation $C_{100\alpha} = M(1 + LSZ\alpha)^{1/L}$, where $Z\alpha$ is the normal equivalent deviate for tail area α . It is also possible to convert measures of baPWV to an exact z -score (standard deviation score) by the equation $z\text{-score} = [Y/M]^L - 1 / [L \times S]$, where Y is the individual baPWV data. The reference percentiles (50th, 75th, 85th, 90th, 95th, and 97th) for age were proposed separately for males and females due to different sex-typical growth patterns.

To test the validity of the baPWV reference values in the assessment of the cardiovascular risk of individuals, we selected 3825 adolescents (2051 males and 1774 females; mean age, 15.1 ± 1.7 years) from 5936 original study participants whose data on body fat mass, WC, and HbA1c were available for analysis (Fig. 1). In this analysis, we divided males and females into three subgroups according to the levels of baPWV z -score normalized for age and compared the subgroups with respect to the anthropometric

Fig. 1 Flow chart of the participant selection procedure. ECG electrocardiography, BMI body mass index, baPWV brachial-ankle pulse wave velocity, TG triglycerides, TC total cholesterol, HDL-C high-density lipoprotein cholesterol, FBG fasting blood glucose, SBP systolic blood pressure, DBP diastolic blood pressure, IFG impaired fasting glucose, WC waist circumference, and HbA1c hemoglobin A1c



and clinical variables as well as the prevalence of metabolic syndrome components, including central obesity (WC: ≥ 90 cm in males and ≥ 80 cm in females; 95th percentile predicted for age and sex in study population), raised BP, raised triglycerides, reduced HDL-C, and impaired fasting glycemia. The cutoff values for these metabolic syndrome components were the same as those mentioned above. In addition, to verify the impact of the atherosclerotic risk factors on baPWV z -score, both males and females were stratified by the clustering number of the metabolic syndrome components, and the baPWV z -scores of the different subcategories were estimated.

The descriptive statistics were mean $\pm$ standard deviation for continuous variables and percentages for categorical variables, unless otherwise stated. The normality of the distribution of data was examined using the Shapiro–Wilk test. If a variable was not normally distributed, then logarithmic transformation was performed before statistical analyses as appropriate. Differences in mean values between sex and among subgroups were analyzed by Student's t test for independent samples, Welch's t test, or an analysis of variance followed by Hochberg's GT2 test for multiple comparisons, whereas differences in frequency were assessed by a chi-squared test. A linear trend test was also performed for the analysis of factors that indicate the dose–response relationship. Correlations of baPWV with anthropometric and clinical variables were analyzed with univariate linear regression, and the variables were tested for their independent predictive effect by multivariate stepwise linear regression. A cutoff of 0.05 was used for the entry and removal of variables in the model.

All statistical analyses were conducted using SPSS statistical package version 22 (IBM/SPSS Inc., Chicago, Illinois, USA). The null hypothesis was rejected at a significance level of $P < 0.05$.

Results

The anthropometric and clinical variables of 4524 healthy adolescents free from atherosclerotic risk factors are summarized in Table 1. Body height, weight, BMI, BMI z -score, SBP, and FBG were significantly higher in males. In contrast, DBP, heart rate, TC, and HDL-C were significantly higher in females. No significant differences were observed between sexes in age and triglycerides.

In this reference population aged between 12 and 18 years, the mean values of baPWV increased from 871 cm/s to 1083 cm/s in males and from 847 cm/s to 988 cm/s in females. Sex-specific differences were present starting from the age of 16 years and revealed significantly higher baPWV values in males than females (909 cm/s in males and 891 cm/s in females aged between 12 and 15 years;

Table 1 Anthropometric and clinical variables of healthy adolescents by sex ($n = 4524$)

Variables	Males ($n = 2272$)	Females ($n = 2252$)	P^a
	Mean $\pm$ s.d	Mean $\pm$ s.d	
Age (yrs)	15.4 $\pm$ 1.7	15.3 $\pm$ 1.8	0.777
Height (cm)	165.5 $\pm$ 9.3	157.5 $\pm$ 6.1	<0.001
Weight (kg)	54.7 $\pm$ 10.7	48.4 $\pm$ 7.2	<0.001
BMI (kg/m ²)	19.8 $\pm$ 2.5	19.4 $\pm$ 2.3	<0.001
BMI z -score ^b	−0.199 $\pm$ 0.939	−0.358 $\pm$ 0.829	<0.001
baPWV (cm/s)	949.3 $\pm$ 123.7	917.6 $\pm$ 112.5	<0.001
SBP (mmHg)	113.0 $\pm$ 9.5	104.0 $\pm$ 7.2	<0.001
DBP (mmHg)	59.2 $\pm$ 6.3	59.6 $\pm$ 6.0	0.024
PP (mmHg)	53.7 $\pm$ 8.1	44.3 $\pm$ 6.4	<0.001
HR (bpm)	64.7 $\pm$ 9.9	67.2 $\pm$ 9.7	<0.001
TG (mg/dl) ^c	54.6 $\pm$ 23.5	54.0 $\pm$ 19.5	0.290
TC (mg/dl)	166.5 $\pm$ 25.9	179.0 $\pm$ 27.7	<0.001
HDL-C (mg/dl)	66.7 $\pm$ 13.2	71.1 $\pm$ 13.5	<0.001
FBG (mg/dl)	88.2 $\pm$ 6.1	87.0 $\pm$ 6.0	<0.001

BMI body mass index, baPWV brachial-ankle pulse wave velocity, SBP systolic blood pressure, DBP diastolic blood pressure, PP pulse pressure, HR heart rate, TG triglycerides, TC total cholesterol, HDL-C high density lipoprotein cholesterol, FBG fasting blood glucose

^aThe P values indicate significant difference by Student's t test or Welch's t test

^bBMI z -score was calculated based on the sex- and age-specific reference values proposed by World Health Organization

^cLogarithmic transformation was performed before statistical analysis

1002 cm/s in males and 957 cm/s in females aged between 16 and 18 years). There was a significant interaction by sex on the association of age with baPWV ($P < 0.001$).

In males, baPWV correlated significantly with age, height, weight, BMI, SBP, DBP, pulse pressure, triglycerides, TC, HDL-C, and FBG (Table 2). In females, baPWV correlated significantly with age, weight, BMI, SBP, DBP, triglycerides, TC, and FBG. After including all variables in baPWV modeling in a stepwise multivariate analysis, for males, age, BMI, SBP, DBP, heart rate, triglycerides, TC, and HDL-C were independent predictors of baPWV, and for females, the independent variables were age, BMI, SBP, DBP, and triglycerides. These models explained 37.2 or 25.5% of the variability seen for baPWV ($P < 0.001$). In the models, age, BMI, SBP, and DBP closely determined the baPWV for both males and females.

The sex-specific L , M , and S values and the smoothed percentiles of baPWV were calculated every half-year for the ages of 12–18 years for males and females (Table 3). The values for L , M , and S were obtained by fitting the cubic splines by nonlinear regression, and the points on each representative percentile curve were calculated using the equation described above. At the age of 18.5 years, the 50th percentile of baPWV was 1086 cm/s for males and

Table 2 Univariate and multivariate regression analysis to evaluate the association of baPWV with anthropometric and clinical variables in healthy adolescents by sex

Variables	Males (<i>n</i> = 2272)				Females (<i>n</i> = 2252)			
	Model 1 ^a		Model 2 ^a		Model 1 ^a		Model 2 ^a	
	β	<i>P</i>	β	<i>P</i>	β	<i>P</i>	β	<i>P</i>
Age (yrs)	0.432	<0.001	0.373	<0.001	0.353	<0.001	0.331	<0.001
Height (cm)	0.218	<0.001			0.011	0.594		
Weight (kg)	0.223	<0.001			0.086	<0.001		
BMI (kg/m ²)	0.168	<0.001	0.149	<0.001	0.109	<0.001	0.086	<0.001
BMI <i>z</i> -score ^b	−0.024	0.247			−0.022	0.302		
SBP (mmHg)	0.455	<0.001	0.261	<0.001	0.307	<0.001	0.161	<0.001
DBP (mmHg)	0.426	<0.001	0.222	<0.001	0.385	<0.001	0.247	<0.001
PP (mmHg)	0.202	<0.001			−0.015	0.471		
HR (bpm)	0.039	0.061	0.044	0.016	0.026	0.227		
TG (mg/dl) ^c	0.143	<0.001	0.038	0.047	0.045	0.045	0.038	0.038
TC (mg/dl)	0.062	0.006	0.094	<0.001	0.051	0.021		
HDL-C (mg/dl)	−0.096	<0.001	−0.085	<0.001	0.015	0.508		
FBG (mg/dl)	−0.054	0.015			−0.055	0.014		

β standardized partial regression coefficient, *BMI* body mass index, *SBP* systolic blood pressure, *DBP* diastolic blood pressure, *PP* pulse pressure, *HR* heart rate, *TG* triglycerides, *TC* total cholesterol, *HDL-C* high-density lipoprotein cholesterol, *FBG* fasting blood glucose

^aModel 1 was univariate regression analysis. Model 2 was multivariate regression analysis with stepwise selection. Values entered into the model were age, height, weight, BMI, BMI *z*-score, SBP, DBP, pulse pressure, HR, TG, TC, HDL-C, and FBG. The goodness of fit of the model (adjusted *R*²) was 0.372 (*P* < 0.001) for males and 0.255 (*P* < 0.001) for females respectively

^bBMI *z*-score was calculated based on the sex- and age-specific reference values proposed by World Health Organization

^cLogarithmic transformation was performed before statistical analysis

987 cm/s for females. These values appear to merge approximately with the baPWV mean values previously reported in healthy adults in their 20 s without any risk factors of atherosclerosis [26]. We proposed the sex-specific 50th, 75th, 85th, 90th, 95th, and 97th percentiles of baPWV normalized for age. Based on the reference percentiles, risk stratification for adolescent males and females can be performed as follows: normal (<85th percentile), mild-to-moderate risk (85th–97th percentile), and high risk (≥97th percentile).

To verify the validity of the sex-specific baPWV reference values, we extracted 3825 adolescents from study participants for further analysis. Among the subgroups of individuals, males and females were grouped according to the baPWV *z*-score, and their anthropometric and clinical variables were compared (Table 4). In males, the subgroup with a *z*-score ≥ 1.0 had higher values for BMI, body fat mass, WC, SBP, DBP, pulse pressure, triglycerides, TC, FBG, and HbA1c and lower levels for HDL-C than those with a *z*-score of <1.0 (*P* for trend <0.05). In females, a significant trend was found with respect to BMI, body fat mass, WC, SBP, DBP, triglycerides, HDL-C, and FBG. In addition, the presence or absence of metabolic syndrome components, including central obesity, raised BP, raised

triglycerides, reduced HDL-C, and impaired fasting glycaemia, were evaluated for the baPWV *z*-score subgroups. The prevalence of all components with the exception of triglycerides for females significantly increased along the baPWV *z*-score level (Table 5). Furthermore, for both males and females, the clustering number of the components showed a graded association with the baPWV *z*-score (−0.08, 0.36, 0.55, 1.26 for males and −0.10, 0.29, 0.52, 1.37 for females) (Fig. 2). Individuals exhibiting three or more of the components manifested the highest value of baPWV *z*-score, and significant differences were observed when compared with the values of those in other subgroups.

Discussion

The present study of healthy Japanese adolescents elucidated that age and sex strongly determined baPWV during adolescence. We analyzed each data set of baPWV in a reference population and constructed sex-specific percentiles normalized to age using the LMS method. The prevalence of metabolic syndrome components, including central obesity, raised BP, raised triglycerides, reduced HDL-C, and impaired fasting glycaemia, were significantly

Table 3 The sex-specific LMS and smoothed percentile values of baPWV according to age in healthy adolescents

Males	Age (years)	<i>n</i>	<i>L</i>	<i>M</i>	<i>S</i>	<i>P</i> 50	<i>P</i> 75	<i>P</i> 85	<i>P</i> 90	<i>P</i> 95	<i>P</i> 97
	12.0	59	0.146	870.1	0.110	871.1	937.1	974.2	1000.1	1039.5	1065.8
	12.5	161	0.230	881.5	0.113	880.0	946.8	984.2	1010.1	1049.6	1075.8
	13.0	254	0.282	889.4	0.115	889.4	957.2	995.1	1021.3	1061.1	1087.6
	13.5	166	0.311	893.6	0.117	899.3	968.5	1007.0	1033.8	1074.3	1101.1
	14.0	179	0.326	912.0	0.118	910.2	980.9	1020.3	1047.6	1089.0	1116.4
	14.5	159	0.338	927.8	0.119	922.1	994.7	1035.1	1063.1	1105.5	1133.7
	15.0	158	0.354	941.6	0.120	935.4	1010.0	1051.6	1080.4	1123.9	1152.8
	15.5	156	0.385	944.9	0.121	950.3	1027.1	1069.9	1099.5	1144.2	1173.8
	16.0	228	0.439	958.1	0.121	967.0	1046.2	1090.2	1120.5	1166.4	1196.7
	16.5	254	0.527	983.9	0.122	985.7	1067.5	1112.6	1143.7	1190.5	1221.3
	17.0	224	0.656	1013.7	0.123	1006.8	1091.2	1137.4	1169.1	1216.6	1247.7
	17.5	156	0.837	1030.7	0.124	1030.4	117.4	1164.5	1196.6	1244.5	1275.7
	18.0	71	1.078	1059.8	0.125	1056.7	1146.3	1194.2	1226.5	1274.3	1305.3
	18.5	47	1.390	1083.0	0.126	1086.1	1178.3	1226.6	1258.9	1306.1	1336.5
Females	Age (years)	<i>n</i>	<i>L</i>	<i>M</i>	<i>S</i>	<i>P</i> 50	<i>P</i> 75	<i>P</i> 85	<i>P</i> 90	<i>P</i> 95	<i>P</i> 97
	12.0	60	0.150	841.1	0.117	841.1	902.5	937.0	960.9	997.4	1021.6
	12.5	148	0.279	855.9	0.120	855.9	919.9	955.6	980.4	1017.9	1042.9
	13.0	258	0.376	869.9	0.121	869.9	935.7	972.4	997.7	1036.0	1061.3
	13.5	164	0.446	883.0	0.123	883.0	950.2	987.5	1013.2	1051.9	1077.5
	14.0	190	0.495	895.5	0.124	895.5	963.7	1001.4	1027.3	1066.4	1092.1
	14.5	186	0.528	907.3	0.125	907.3	976.3	1014.3	1040.5	1079.8	1105.7
	15.0	200	0.550	918.5	0.126	918.5	988.4	1026.8	1053.2	1092.8	1118.9
	15.5	150	0.565	929.2	0.126	929.2	1000.1	1039.1	1065.8	1105.9	1132.3
	16.0	214	0.580	939.5	0.127	939.5	1011.9	1051.6	1078.8	1119.7	1146.5
	16.5	182	0.599	949.4	0.127	949.4	1023.8	1064.6	1092.6	1134.4	1162.2
	17.0	226	0.627	959.0	0.127	959.0	1036.3	1078.7	1107.8	1151.4	1180.0
	17.5	100	0.670	968.4	0.128	968.4	1049.7	1094.2	1124.8	1170.6	1200.7
	18.0	77	0.733	977.7	0.128	977.7	1064.1	1111.5	1144.1	1192.8	1224.8
	18.5	97	0.821	986.8	0.129	986.8	1080.0	1131.1	1166.2	1218.7	1253.2

L skewness, *M* median, *S* coefficient of variation, *P* percentile

correlated with the levels of standardized baPWV *z*-score for both males and females. Thus, the provided reference values of baPWV may contribute to a more precise classification of vascular abnormalities in adolescents and could help determine individuals with cardiovascular risk, who should thereby be subjected to detailed observation and for whom several health management treatments should be considered.

Arterial stiffness can be assessed by PWV, which reflects alterations of structural and functional properties of the arterial tree [7]. While carotid-to-femoral PWV (cfPWV) has been regarded as an established and reliable measure for arterial stiffness [27, 28], a simple and convenient oscillometric device for the measurement of baPWV has recently been introduced for clinical use [10]. baPWV has an advantage in clinical research studies because its measurement is simple and noninvasive, does not require special skills, and is relatively free from operator bias compared

with other vascular assessments [10, 11]. Thus, baPWV measurements can be applied for the assessment of early vascular changes in children and adolescents. The validity and reproducibility of baPWV and its potential use for screening subclinical organ damage in a large population have already been demonstrated [10–16]. Furthermore, it has been shown that baPWV is closely correlated with the directly measured aortic PWV, and its ability to predict the presence of both stroke and coronary artery disease was comparable to that of cfPWV [29]. Furthermore, a meta-analysis using individual participant data clearly established baPWV as an independent predictor for the risk of developing cardiovascular disease in low- to intermediate-risk populations [17].

Corroborating the findings of other studies in children and adolescents [2, 18, 20, 22], the baPWV of study participants showed a continuous progressive increase for both males and females. The stiffness of the arterial wall is an

Table 4 Anthropometric and clinical variables among subgroups stratified by the baPWV *z*-score in study participants by sex (*n* = 3825)

	baPWV <i>z</i> -score			<i>P</i> ^a	
	<−1.0	−1.0 to +1.0	≥+1.0		
	Mean ± s.d	Mean ± s.d	Mean ± s.d	ANOVA	Trend
Males					
<i>n</i>	320	1366	365		
BMI (kg/m ²)	20.1 ± 3.2	20.7 ± 3.2	21.1 ± 4.1	0.004	0.001
BMI <i>z</i> -score ^b	-0.150 ± 1.127	0.039 ± 0.998	0.062 ± 1.148	0.009	0.011
Body fat mass (%)	17.3 ± 6.1	18.9 ± 5.9	19.9 ± 6.3	<0.001	<0.001
WC (cm)	69.6 ± 8.7	71.5 ± 8.9	72.1 ± 10.7	0.001	0.001
SBP (mmHg)	109.3 ± 9.6	115.3 ± 9.7	121.9 ± 11.2	<0.001	<0.001
DBP (mmHg)	55.9 ± 6.2	60.4 ± 7.1	64.9 ± 7.7	<0.001	<0.001
PP (mmHg)	53.4 ± 7.9	54.9 ± 8.5	57.0 ± 8.6	<0.001	<0.001
TG (mg/dl) ^c	54.1 ± 26.4	62.6 ± 35.6	73.2 ± 48.0	<0.001	<0.001
TC (mg/dl)	161.7 ± 25.9	168.6 ± 27.9	172.5 ± 26.8	<0.001	<0.001
HDL-C (mg/dl)	70.2 ± 14.4	66.2 ± 14.6	63.3 ± 14.4	<0.001	<0.001
FBG (mg/dl)	87.4 ± 7.6	89.1 ± 6.9	90.0 ± 8.7	<0.001	<0.001
HbA1c (%)	5.10 ± 0.32	5.16 ± 0.29	5.20 ± 0.35	0.002	<0.001
Females					
<i>n</i>	294	1158	322		
BMI (kg/m ²)	19.8 ± 2.5	20.1 ± 2.5	20.4 ± 3.1	0.006	0.001
BMI <i>z</i> -score ^b	−0.238 ± 0.840	−0.182 ± 0.857	−0.037 ± 0.931	0.009	0.004
Body fat mass (%)	24.1 ± 5.4	24.7 ± 5.5	25.7 ± 6.0	0.003	0.001
WC (cm)	67.5 ± 6.4	67.9 ± 6.7	69.7 ± 7.2	<0.001	<0.001
SBP (mmHg)	101.8 ± 7.6	105.1 ± 7.8	110.3 ± 8.7	<0.001	<0.001
DBP (mmHg)	56.7 ± 5.8	60.6 ± 6.5	67.9 ± 7.4	<0.001	<0.001
PP (mmHg)	45.1 ± 6.8	44.5 ± 7.1	45.4 ± 7.2	0.079	0.489
TG (mg/dl) ^c	57.9 ± 27.9	62.1 ± 27.7	62.5 ± 25.9	0.015	0.011
TC (mg/dl)	181.9 ± 27.3	180.4 ± 29.2	177.8 ± 27.3	0.185	0.073
HDL-C (mg/dl)	73.6 ± 15.2	71.6 ± 14.6	69.0 ± 15.9	0.001	<0.001
FBG (mg/dl)	87.0 ± 6.4	87.4 ± 7.1	88.8 ± 7.3	0.002	0.002
HbA1c (%)	5.19 ± 0.25	5.17 ± 0.28	5.19 ± 0.27	0.531	0.516

BMI body mass index, WC waist circumference, SBP systolic blood pressure, DBP diastolic blood pressure, PP pulse pressure, TG triglycerides, TC total cholesterol, HDL-C high-density lipoprotein cholesterol, FBG fasting blood glucose, HbA1c hemoglobin A1c

^aThe *P* values indicate significant difference by one-way ANOVA or trend test

^bBMI *z*-score was calculated based on the sex- and age-specific reference values proposed by World Health Organization

^cLogarithmic transformation was performed before statistical analysis

important determinant of vascular impedance that determines the buffer and conduit functions of arteries as well as ventricular vascular coupling, which is considered a key pathophysiological mechanism in several cardiovascular diseases [30]. Thus, early assessment of the loss of arterial wall elasticity provides important information regarding the subclinical vascular damage. However, the clinical application of baPWV as a marker of arterial stiffness in pediatric populations should consider the physiological development characteristics of children and adolescents [2, 18, 22, 23]. In childhood, as characterized by age-related

developmental processes, a greater stroke volume in response to the development of cardiac function may increase the arterial pressure and affect the baPWV [31]. Furthermore, arteries undergo significant changes, including a thickening of the muscular medial layer related to progressively increasing BP and a reduction of vascular compliance and distensibility associated with early medial restructuring and increased deposition of collagen [32, 33].

The baPWV of study participants was higher in males than in females, and the difference became more pronounced with the increasing age, which is consistent with

Table 5 Prevalence of metabolic syndrome components among subgroups stratified by the baPWV *z*-score in study participants

	Males				<i>P</i> ^a		Females				<i>P</i> ^a	
	<−1.0	−1.0 to +1.0	≥+1.0				<−1.0	−1.0 to +1.0	≥+1.0			
	<i>n</i> (%)	<i>n</i> (%)	<i>n</i> (%)		χ^2	Trend	<i>n</i> (%)	<i>n</i> (%)	<i>n</i> (%)		χ^2	Trend
Central obesity ^b	9 (2.8)	62 (4.5)	30 (8.2)		0.003	0.001	10 (3.4)	67 (5.8)	37 (11.5)		<0.001	<0.001
Raised BP ^b	4 (1.3)	104 (7.6)	86 (23.6)		<0.001	<0.001	9 (3.1)	101 (8.7)	85 (26.4)		<0.001	<0.001
Raised TG ^b	7 (2.2)	86 (6.3)	24 (6.6)		0.013	0.017	10 (3.4)	76 (6.6)	21 (6.5)		0.116	0.115
Reduced HDL-C ^b	8 (2.5)	68 (5.0)	28 (7.7)		0.008	0.002	13 (4.4)	58 (5.0)	30 (9.3)		0.008	0.007
IFG ^b	13 (4.1)	64 (4.7)	37 (10.1)		<0.001	<0.001	14 (4.8)	63 (5.4)	38 (11.8)		<0.001	<0.001

BP blood pressure, TG triglycerides, HDL-C high-density lipoprotein cholesterol, IFG impaired fasting glycemia

^aThe *P* values indicate significant difference by chi-square test or trend test

^bParticipants who had more than or equal to the sex- and age-specific 95th percentile value for waist circumference, BP, TG, and FBG (less than the 5th percentile value for HDL-C) were defined as those with central obesity, raised BP, raised TG, reduced HDL-C, and raised FBG respectively

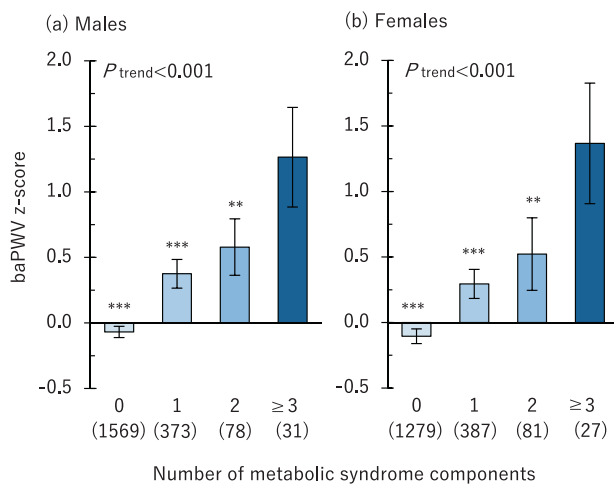


Fig. 2 The baPWV *z*-scores among subgroups stratified by clustering number of metabolic syndrome components in study participants (*n* = 3825). Data are shown as the mean (95% confidence interval). Values in parentheses are numbers of participants. The *P* values indicate statistical significance by trend test. ***P* < 0.01, ****P* < 0.001 vs. subgroup having three or more metabolic syndrome components (Hochberg's GT2 test)

previous studies [22, 23, 34]. The relatively low stroke volume and arterial pressure presumably account for an appreciable part of the more gradual increase of baPWV in females [31]. However, from another point of view, it is possible that the beneficial effect of female sex hormones is the reason for the sex differences observed in baPWV during adolescence. Female sex hormones are believed to work protectively against the increase in arterial stiffening in postpubertal females [16, 26, 35]. In relation to this, our study revealed that metabolic parameters, including triglycerides, TC, and HbA1c, were significantly correlated with the baPWV *z*-score in males but not in females. Although a plausible explanation for the lack of correlations in females

is not entirely demonstrated, more excretion of estrogen may also contribute to the sex-based difference by providing vascular protection together with improving lipid metabolism and glucose tolerance [36, 37]. We could not present data on the pubertal status of individuals; however, as the study participants were aged over 12 years, almost all the females were assumed to be postpubertal. Thus, the elevated levels of estrogen in these females might have altered the impact of risk factors on baPWV.

As outlined above, our findings support the suggestion that it is important to consider a participant's age and sex when classifying vascular abnormalities of adolescents that might be caused by the presence of risk factors for atherosclerosis. In this study, the sex-specific physiological values of baPWV controlling for age were proposed for a reference adolescent population free from several risk factors. Individuals with risk factors who were excluded for this analysis had significantly higher baPWV values than the reference group in both males (1013 cm/s vs. 949 cm/s, *P* < 0.001) and females (955 cm/s vs. 918 cm/s, *P* < 0.001). The reference percentiles of baPWV were generated using the LMS method [25]. The LMS method provides a way of obtaining growth percentile standards and allows the calculation of smoothed percentiles and accurately normalized *z*-scores. This method has recently been applied to the description of not only pediatric anthropometric data but also clinical measurements, including arterial stiffness parameters [31, 34, 38]. Furthermore, LMS values and standard percentiles were determined using data from a much larger sample of healthy adolescents than in previous studies [22, 23]. In addition, the standardized 50th percentile values of baPWV at the age of 18.5 years for males and females in our participants appeared to merge into the values of those in their 20s, as previously reported in a study of a healthy adult population without individuals exhibiting risk factors for atherosclerosis [26]. Research of

this kind is scarce for children and adolescents; therefore, the reference values of baPWV proposed in this study are potentially useful in clinical practice and population studies for determining cardiovascular risk profiles.

To confirm the validity of the baPWV reference values for the assessment of vascular abnormalities in an adolescent population, the baPWV *z*-score subgroups were compared with respect to the presence or absence of central obesity, raised BP, raised triglycerides, reduced HDL-C, and impaired fasting glycemia. The results highlighted that the prevalence of all components with the exception of triglycerides for females significantly increased along the levels of the standardized increase in the baPWV *z*-score. We also demonstrated a cumulative relationship between an increasing number of metabolic syndrome components and baPWV *z*-score values. Therefore, the use of baPWV standards could help eliminate the impact of age and sex and permit a more precise classification of arterial stiffness during adolescence. Furthermore, these data suggest that, in addition to the development of the cardiovascular system that is accompanied by an increase in arterial pressure, the presence or accumulation of risk factors contributes to the acceleration of baPWV in adolescents. In accordance with these findings, some researchers have shown positive associations between atherosclerotic changes and risk factors such as obesity, elevated BP, lipid and glucose tolerance abnormalities, total homocysteine levels, and parental hypertension in a group of healthy children and adolescents [2, 18–21, 23, 39]. The observations in the present and previous studies indicate that arterial wall structure and function can be influenced by atherosclerotic risk factors as early as the second decade of life, and the vascular changes may not be confined to individuals suffering from severe obesity, hypertension, hyperlipidemia, or diabetes mellitus. This fact emphasizes not only the importance of strategies to control risk factors for atherosclerosis but also the need to evaluate the arterial wall properties of adolescents in the early stage of vascular aging when clinically detectable cardiovascular diseases are rarely seen.

We have also proposed baPWV level classifications as normal, mild-to-moderate risk, and high risk on the basis of the smoothed percentile values in standardized curves. To use the classification table in a clinical setting, the participant's measured baPWV is compared with the table for half-year ages. An individual with a baPWV below the 85th percentile (mean + 1 standard deviation) is classified as being at low risk for cardiovascular disease. If the participant's baPWV is above the 97th percentile (mean + 2 standard deviation), the individual is classified as being at high risk for cardiovascular disease. A baPWV between the 85th and 97th percentiles is mild-to-moderate and warrants further observation and the evaluation of other risk factors. A more precise characterization of an individual's baPWV

level is an average of multiple measurements taken for weeks to months. Subsequently, when baPWV is confirmed to exceed the 97th percentile reference cutoff value by repeated measurements, further observation and several treatments should be considered.

Several potential limitations of this study should be considered. First, we proposed reference percentile values of baPWV using data from a large cohort of adolescents compared with data from previous studies [22, 23]. However, the study participants included only Japanese adolescents from the West Japan region. In addition, we did not confirm whether the participants were representative of the overall Japanese adolescent population. Thus, this potentially limits the ability to extrapolate the results of this study to other populations. Second, our data revealed close associations between baPWV and risk factors that may contribute to the acceleration of arterial stiffening. Thus, baPWV has the potential to be a useful indicator for identifying individuals with early vascular abnormalities in adolescents. However, we did not compare the validity of baPWV with other available vascular assessments. Third, because increased arterial stiffening is an important cause of atherosclerosis in adulthood, baPWV measurements in adolescents have the potential to provide additional predictive information on the cardiovascular risk of individuals over that obtained from traditional risk factors. However, further longitudinal studies are needed to assess the relationship between higher baPWV values during adolescence and the risk of cardiovascular events in adulthood.

In conclusion, the observations in the present study indicate that baPWV measurements are feasible for the assessment of arterial wall properties of individuals during adolescence. However, some adjustments related to age and sex are required to determine the vascular abnormalities on the basis of baPWV. This is the first study providing age- and sex-specific physiological values for baPWV in healthy Japanese adolescents. The reference percentiles and their derived standardized baPWV *z*-scores are closely related to the presence or absence of atherosclerotic risk factors. The data could help to classify the extent of vascular abnormalities and may thereby contribute to the identification of individuals at risk for cardiovascular disease in later life.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflicts of interest.

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